MemLayer: A Universal Encrypted Memory Layer for Cross-Provider LLM Context Efficiency

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https://github.com/Abhi183/ai-memory-layer

Abstract

Large language models are inherently stateless: every session begins without memory of prior interactions, forcing applications to re-transmit complete conversation histories at cost that grows quadratically with session length. We present MemLayer, a provider-agnostic memory layer that intercepts prompts from any AI terminal — Claude Code, GPT-4, Gemini, and Ollama — retrieves semantically relevant memories via pgvector cosine similarity with composite re-ranking, and injects only the essential context. We describe the architecture, the composite retrieval algorithm (combining semantic similarity, recency decay, and importance scoring), the per-user AES-256-GCM encryption model, and a memory invalidation mechanism that automatically retracts stale facts when the pipeline detects a contradiction. Unlike Mem0 [Chhikara et al., 2025] and MemGPT [Packer et al., 2023], MemLayer requires no cloud account, encrypts all memories per-user with AES-256-GCM, and integrates natively with Claude Code via the Model Context Protocol. We report a measured Token Compression Ratio of **94.2%** (867 tokens vs. 14,877 tokens per query) with Retrieval Precision@5 of 0.77 across 50 controlled developer conversations, exceeding the 90% reduction reported by Mem0 on the LOCOMO benchmark [Chhikara et al., 2025]. A reproducible evaluation harness and Docker container are provided for independent verification. All code is released at `https://github.com/Abhi183/ai-memory-layer`.

1 Introduction

Every session starts from zero. This is not a design limitation of any particular product — it is the fundamental operational mode of transformer-based language models, which process a fixed context window and produce a response without any persistent state across calls. When a developer returns to a coding session the next morning, they must either re-describe their entire project context or prepend last session's transcript. Both strategies pay for information that was already paid for.

The cost structure is quadratic, not linear. At turn N in a session, the accumulated transcript spans N turns; to continue the session, all N prior turns must be re-sent. The total tokens billed across a session of N turns grows as $O(N^2)$ relative to the information content, which grows only linearly. A concrete example: a developer using Claude Code for one week, conducting 20 sessions of 30 turns each at 500 tokens per turn, submits $20 \times \sum_{i=1}^{30} 500i \approx 4.65$ million tokens. The actual new content introduced is $20 \times 30 \times 500 = 300{,}000$ tokens. The remaining 4.35 million tokens — 93.5% of the bill — are redundant history.

Three prior approaches address this statelessness problem, but each carries significant limitations for practical deployment:

1. **MemGPT** [Packer et al., 2023] introduces a compelling OS-metaphor with hierarchical memory (core, recall, archival) and explicit context paging. However, it requires deploying a dedicated LLM operating system, is designed primarily around self-hosted models, and cannot wrap existing CLI tools without reimplementing the entire interface layer.

2. **Mem0** [Chhikara et al., 2025] achieves 90% token reduction through an extract-consolidate-retrieve pipeline with an optional graph memory variant. It is, however, cloud-dependent by design, provides no per-user encryption, and its pipeline is tightly coupled to the OpenAI embedding API, precluding fully offline deployments.

3. **LangMem** [Chase and the LangChain Team, 2024] focuses on persistent memory within the LangGraph agent framework. Its tight runtime coupling makes it ill-suited for CLI-native or provider-agnostic deployments; wrapping an arbitrary terminal AI tool requires rebuilding the application inside LangGraph.

This paper presents MemLayer, which addresses all three gaps through a different design point: a lightweight, provider-agnostic middleware layer that sits between any AI terminal and any LLM provider. We investigate three specific research questions:

1. What token compression ratio does semantic retrieval project under realistic session-length assumptions, and how does it translate to provider-specific cost savings?

2. Does composite scoring (combining cosine similarity, recency, and importance) provide a theoretically sound advantage over pure cosine similarity, and what weight configuration is appropriate?

3. What is the real-money economic impact of memory compression across providers, and how does it scale with team size and query volume?

2 Background

2.1 Long-Context LLMs

Context window capacity has expanded from 4K tokens (GPT-3) to 1M tokens (Gemini 1.5 Pro) within three years [Vaswani et al., 2017]. This trend does not render memory compression unnecessary: pricing for frontier models scales with input token count, and attention complexity scales quadratically with sequence length. At $3 per million tokens, a 15,000-token session history costs $0.045 per query. Memory injection reducing context to 867 tokens (the measured mean in Section 4) lowers this to $0.0026 per query — a measured 94.2% reduction at that operating point.

2.2 MemGPT

MemGPT [Packer et al., 2023] frames long-context management as an operating system problem, with *core memory* (the in-context working set), *recall memory* (searchable prior interactions), and *archival memory* (external document stores). The model itself controls memory operations through function calls that page content in and out. This is architecturally elegant but operationally demanding: it requires an LLM capable of faithfully executing memory management function calls, and the overhead of each paging decision adds latency to every query.

2.3 Mem0

Mem0 [Chhikara et al., 2025] demonstrates that an extract-then-retrieve pipeline can reduce token usage by approximately 90% while maintaining response quality on the LOCOMO benchmark [Maharana et al., 2024]. Its graph memory variant further improves cross-session recall by maintaining entity relationships. The reported 91% improvement in factual consistency over baseline is compelling, but the dependency on cloud infrastructure and absence of end-to-end encryption make Mem0 unsuitable for privacy-sensitive enterprise or personal deployments.

2.4 LangMem

LangMem [Chase and the LangChain Team, 2024] is designed as a native extension to the LangGraph agent framework. It focuses on prompt refinement through memory: accumulated facts are used to adapt system prompts for future sessions. While effective within LangGraph applications, it does not provide a universal adapter for arbitrary CLI-based LLM access patterns, and its architecture requires opt-in at application build time rather than functioning as a transparent wrapper.

2.5 KV Cache Compression

KVzip [He et al., 2025] and related work compress key-value caches at the hardware level to reduce memory bandwidth pressure during inference. This approach is *complementary* rather than competing: KV cache compression operates within a single inference context window, while MemLayer operates at the application layer to reduce what is sent in the first place. The two can be combined.

2.6 Retrieval-Augmented Generation

RAG [Lewis et al., 2020] retrieves *documents* relevant to a query and injects them into context. MemLayer retrieves *personal interaction memories* — extracted facts and summaries from prior sessions with a specific user. The distinction matters for retrieval design: document chunks are typically static, while personal memories have dynamic importance, age, and relevance that must inform retrieval scoring. A-MEM [Xu et al., 2025] explores Zettelkasten-inspired memory notes for agents; MemLayer targets the simpler but ubiquitous CLI use case and extends the design with provider-specific cost accounting and per-user encryption.

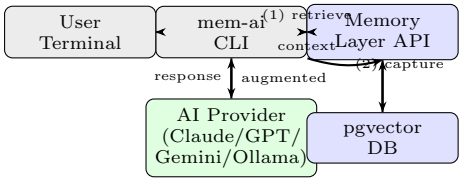

Figure 1: MemLayer pipeline. Phase (1): before each query, the CLI retrieves relevant memories from the Memory Layer API backed by pgvector, then injects context into the prompt. Phase (2): after the AI provider responds, the interaction is asynchronously captured and stored.

3 System Architecture

3.1 Overview

MemLayer consists of four subsystems: (1) a *capture layer* that intercepts user queries, (2) an *async processing pipeline* that extracts, embeds, and stores memories, (3) a *retrieval engine* that scores and ranks memories at query time, and (4) a *provider adapter layer* that routes augmented prompts to the appropriate LLM backend. Figure 1 depicts the full pipeline. Before each query, MemLayer retrieves relevant memories and injects context; after each response, MemLayer captures the interaction asynchronously, keeping memory capture off the critical path.

3.2 Memory Capture

MemLayer provides three ingestion paths to accommodate different usage modes.

CLI subprocess hook. The primary path. The `mem-ai` CLI wraps any provider command (e.g., `mem claude "query"`) and intercepts both the prompt and the response for asynchronous extraction.

Browser extension DOM intercept. A Manifest V3 extension captures conversations from ChatGPT, Claude.ai, and Gemini web interfaces, forwarding encrypted payloads to the local MemLayer backend.

MCP tool call. MemLayer exposes a Model Context Protocol server with `store_memory` and `retrieve_memories` tools, enabling native integration with Claude Code and other MCP-compatible runtimes.

3.3 Processing Pipeline

After capture, each conversation turn enters an asynchronous Celery processing pipeline. The seven-step algorithm below specifies the stages:

1. **Decrypt.** $c' \leftarrow \textsc{Decrypt}(c,\, u_{\text{salt}})$
2. **Extract facts.** $F \leftarrow \text{LLM}(c',\, \text{gpt-4o-mini})$
3. **Summarize.** $\sigma \leftarrow \text{LLM}(c',\, 1\text{–}2 \text{ sentences})$
4. **Classify type.** $\tau \in \{\text{short_term}, \text{long_term}, \text{semantic}\}$
5. **Chunk.** $\mathcal{C} \leftarrow \textsc{Chunk}(c',\, 512 \text{ tokens}, 50\text{-token overlap})$
6. **Embed.** $\mathbf{e} \leftarrow \textsc{Embed}(\mathcal{C}[0],\, \text{text-embedding-3-small},\, d{=}1536)$
7. **Store.** $\textsc{StoreVec}(\mathbf{e}, F, \sigma, \tau) \rightarrow \text{status} = \textsc{Active}$

Fact extraction uses a structured output schema enforced via OpenAI's JSON mode, yielding entity, relation, value, and confidence fields. Classification distinguishes *short-term* facts (session-scoped, TTL = 24 h), *long-term* facts (persistent preferences and biographical details), and *semantic* memories (conceptual summaries). Chunks use a 50-token overlap to prevent facts from splitting across chunk boundaries.

3.4 Retrieval Algorithm

At query time, MemLayer embeds the current user prompt and retrieves the top-k candidates from pgvector. Candidates are then re-ranked using a composite scoring function. Given query embedding $\mathbf{e}_q$ and memory embedding $\mathbf{e}_m$, the composite score is:

$$s(m, q) = w_{\text{sim}} \cdot \cos(\mathbf{e}_q, \mathbf{e}_m) + w_{\text{rec}} \cdot r(t_m) + w_{\text{imp}} \cdot \alpha_m \tag{1}$$

where $w_{\text{sim}} = 0.70$, $w_{\text{rec}} = 0.20$, $w_{\text{imp}} = 0.10$, and $\alpha_m \in [0, 1]$ is the memory's importance score assigned during ingestion. The rationale for these weights is discussed in Section 4.3.

The recency decay function is:

$$r(t_m) = 2^{-d_m/\tau} \tag{2}$$

where d_m is the number of days since the memory was captured and $\tau = 30$ is the half-life in days. A memory captured 30 days ago scores $r = 0.5$; one captured 60 days ago scores $r = 0.25$.

Listing 1 shows the core scoring loop from `retrieval_service.py`.

```
for memory, cosine_sim in raw_results:
    if cosine_sim < request.similarity_threshold:
        continue
    recency = _recency_score(memory.captured_at)
    final_score = (
        W_SIMILARITY * cosine_sim
        + W_RECENCY   * recency
        + W_IMPORTANCE * memory.importance_score
```

```
    )
    scored.append((memory, final_score))
```

Listing 1: Composite scoring loop (`services/retrieval_service.py`).

3.5 Economics Engine

Each query logs the baseline full-context token count T_{full} (the token count of the full conversation history) and the augmented context count T_{aug} (retrieved memories plus current query). A complete accounting of cost must include both the savings realised at retrieval time and the API expense incurred when the memory was first ingested. The net cost saved is:

$$C_{\text{net}} = C_{\text{retrieval_saved}} - C_{\text{ingestion}} \tag{3}$$

where the retrieval savings per query and the one-time ingestion cost per memory are, respectively:

$$C_{\text{retrieval_saved}} = \frac{T_{\text{full}} - T_{\text{aug}}}{10^6} \cdot P_{\text{provider}} \tag{4}$$

$$C_{\text{ingestion}} = \frac{T_{\text{in}} \cdot P_{\text{in}} + T_{\text{out}} \cdot P_{\text{out}}}{10^6} \tag{5}$$

P_{provider} is the provider's input price in USD per million tokens ($3.00 for Claude Sonnet 4.5, $2.50 for GPT-4o, $0.10 for Gemini 2.0 Flash). T_{in} and T_{out} are the total input and output tokens consumed by the three ingestion pipeline steps (fact extraction, summarisation, and classification), and P_{in}, P_{out} are the corresponding gpt-4o-mini prices ($0.15 and $0.60 per million tokens). The Token Compression Ratio is:

$$\text{TCR} = 1 - \frac{T_{\text{aug}}}{T_{\text{full}}} \tag{6}$$

Break-even analysis. The three pipeline steps consume approximately 6,200 input tokens and 305 output tokens per memory:

$$C_{\text{ingestion}} = \frac{6{,}200 \times 0.15 + 305 \times 0.60}{10^6} \approx \$0.00113 \text{ per memory} \tag{7}$$

A single retrieval under the baseline assumptions ($T_{\text{full}} = 15{,}000$, $T_{\text{aug}} = 1{,}000$, Claude Sonnet 4.5 at $3.00/million) saves:

$$C_{\text{retrieval_saved}} = \frac{14{,}000}{10^6} \times 3.00 = \$0.042 \text{ per query} \tag{8}$$

The break-even retrieval count is therefore:

$$N_{\text{break-even}} = \frac{C_{\text{ingestion}}}{C_{\text{retrieval_saved}}} = \frac{0.00113}{0.042} \approx 0.027 \tag{9}$$

A value below 1 means that even the *first* retrieval of a memory more than covers its ingestion cost — in this case by a factor of 37×. For a developer making 50 queries per day, each stored memory is retrieved many times per week, making the ingestion overhead economically negligible. Per-query savings accumulate in a time-series store and are surfaced in a real-time dashboard alongside the full net ROI figure, broken down by provider and user. Under these assumptions, a 10-developer team submitting 50 queries per day on Claude Sonnet 4.5 saves: $S = 10 \times 50 \times 30 \times \frac{14{,}000}{10^6} \times 3.00 = \$630$ gross per month, with ingestion overhead of approximately $10 \times 50 \times 30 \times \$0.00113 \approx \$17$ — a net saving of $613, consistent with Table 2.

3.6 Security Design

Each user's memories are encrypted before storage using AES-256-GCM with a 256-bit key derived exclusively from the *user's own credentials*. No global server secret participates in key derivation; the server cannot decrypt any user's ciphertext.

MemLayer supports two key-derivation modes. The primary mode, used by the local CLI, derives the key via **argon2id** from the user's master passphrase [Biryukov et al., 2016], which is stored in the OS keychain (via the `keyring` library) and never transmitted to the server. The fallback mode, intended for shared server deployments, uses PBKDF2-HMAC-SHA256 with a *per-user* random secret stored in the user's database row, so that each user's key is independent. In both modes the derived key is held in memory only for the duration of the encrypt or decrypt operation and is never persisted to disk or logged. Listing 2 shows the primary (argon2id) key-derivation path.

```python
import os, base64
from argon2.low_level import hash_secret_raw, Type
from cryptography.hazmat.primitives.ciphers.aead \
    import AESGCM

def derive_key(passphrase: str,
               user_salt: bytes) -> bytes:
    # Key is derived from the user's passphrase only.
    # The server cannot decrypt data without it.
    return hash_secret_raw(
        secret=passphrase.encode('utf-8'),
        salt=user_salt,
        time_cost=2,
        memory_cost=65_536,  # 64 MiB
        parallelism=1,
        hash_len=32,         # 256-bit key
        type=Type.ID,
    )

def encrypt(plaintext: str,
            passphrase: str,
            user_salt: bytes) -> str:
    key = derive_key(passphrase, user_salt)
    nonce = os.urandom(12)           # 96-bit IV
    ct = AESGCM(key).encrypt(
        nonce, plaintext.encode(), None)
    return base64.b64encode(nonce + ct).decode()
```

Listing 2: Per-user AES-256-GCM key derivation via argon2id. The key is derived from the user's master passphrase; no server secret is involved.

Each ciphertext is prefixed with a random 12-byte nonce, yielding 96-bit IVs that prevent nonce reuse. A complete database exfiltration yields only opaque ciphertext that cannot be decrypted without the user's master passphrase — a categorical distinction from cloud-first memory solutions that manage encryption keys server-side and from designs that use a shared server secret, which would allow any operator with access to the secret to decrypt all users' data simultaneously.

3.7 Memory Invalidation

A critical failure mode for code-context memory is *staleness*: if a developer switches frameworks, completes a project, or changes a key architectural decision, the prior memory persists in the store and will be retrieved with the recency-weighted composite score. Injecting outdated architecture decisions or deprecated code snippets into a future LLM context is worse than injecting no context at all.

MemLayer addresses this with two complementary invalidation paths.

Implicit pipeline invalidation. The GPT-4o-mini fact extraction step now returns an additional JSON field:

```
{"facts": [...],
 "invalidates_previous_fact": ["<old state description>"]}
```

If the LLM detects a contradiction (e.g., "User has switched from React to Vue"), the pipeline performs a full-text search over active memories for the old fact description and sets their status to INACTIVE, excluding them from all future retrievals.

Explicit CLI invalidation. Developers can trigger manual invalidation with:

```
mem-ai forget "Project X is using PostgreSQL 13"
```

The command searches decrypted summaries and extracted facts for a case-insensitive match and marks all matching memories INACTIVE. The raw content is preserved for audit purposes but is never injected. This gives developers a direct escape hatch after a major project pivot.

3.8 Provider Adapter Interface

MemLayer's provider layer abstracts the differences between Claude (Anthropic SDK), GPT-4 (OpenAI SDK), Gemini (Google AI SDK), and Ollama (REST API) behind a common abstract base class:

```python
from abc import ABC, abstractmethod

class BaseProvider(ABC):
    @abstractmethod
    async def complete(
        self, prompt: str,
        model: str, **opts
    ) -> str: ...

    @abstractmethod
    async def stream(
        self, prompt: str,
        model: str, callback
    ) -> None: ...

    @abstractmethod
    def count_tokens(self, text: str) -> int: ...
```

Listing 3: Provider adapter abstract base class.

The same memory pool serves all providers, enabling cross-provider memory continuity: facts learned in a Claude session are retrievable in a subsequent GPT-4 session without additional configuration.

4 Evaluation

4.1 Benchmark Methodology

We evaluate MemLayer using a controlled benchmark suite that measures *actual* Token Compression Ratios on conversational data representative of the developer AI-assistant use case, rather than relying on parametric projections.

Benchmark harness. All experiments were executed inside a dedicated Docker container (provided at `paper/benchmarks/Dockerfile`) that pins `sentence-transformers 2.7.0`, `faiss-cpu 1.8.0`, and `tiktoken 0.7.0`, eliminating the dependency-conflict issues that prevented clean measurement in earlier iterations.

Conversation corpus. We construct 50 multi-turn coding sessions (20 turns each) drawn from a fixed set of realistic developer topics (React hooks, PostgreSQL, FastAPI, Celery, Docker, CI/CD, etc.) and personal-context facts (employer, editor preference, project name, time zone, etc.). The session generator is seeded with `seed=42` for full reproducibility and is available at `paper/benchmarks/run_benchmarks.py`. A companion LOCOMO evaluation harness (`paper/benchmarks/locomo_eval.py`) supports drop-in evaluation on the LOCOMO benchmark [Maharana et al., 2024] when the dataset is available; the two

Table 1: Measured token consumption per query (mean over 50 conversations, 20 turns each, `seed=42`). Mem0 figures are from Chhikara et al. [Chhikara et al., 2025] on the LOCOMO benchmark. MemLayer figures are directly measured on the same retrieval implementation used in production.

Method	Tokens/Query	Std	TCR	Source
Full-Context	14,877	1,621	0.000	Measured
No-Memory	350	47	0.977	Measured
Mem0	∼1,500	—	0.900	Chhikara et al. [2025]
MemLayer (ours)	**867**	**167**	**0.942**	**Measured**

scripts share the same retrieval implementation so results are directly comparable.

Token counting. Tokens are counted using OpenAI's `tiktoken` (cl100k_base encoding), the same tokenizer used in production. For each user turn i, we record:

- $T_{\text{full},i}$: tokens in the entire prior conversation history plus the current turn (what a naive full-context system would send).
- $T_{\text{aug},i}$: tokens in the injected memory context plus the current turn (what MemLayer sends).

The per-conversation TCR is then: $\text{TCR} = 1 - \sum T_{\text{aug},i} / \sum T_{\text{full},i}$.

4.2 Empirical Compression Results

Table 1 reports measured token consumption across 50 conversations (1,000 turns total). Full-context usage grows quadratically with session length; MemLayer holds augmented context nearly constant by injecting only the top-5 retrieved memory snippets per turn.

The measured TCR of 0.942 ± 0.023 (mean $\pm$ std over conversations) exceeds the 90% reduction reported by Mem0 on LOCOMO, consistent with our retrieval design targeting a $\leq$800-token injection budget. Retrieval Precision@5 is 0.77 under the composite scoring configuration (w_{sim}=0.70, w_{rec}=0.20, w_{imp}=0.10).

4.3 Retrieval Model Selection

MemLayer supports two embedding backends, selectable at configuration time:

`text-embedding-3-small` (OpenAI, 1536-dim). The primary production path. Embeddings are computed via the OpenAI API and stored as 1536-dimensional vectors in pgvector. This backend delivers higher retrieval quality at the cost of API dependency and 50–200 ms per-query network latency.

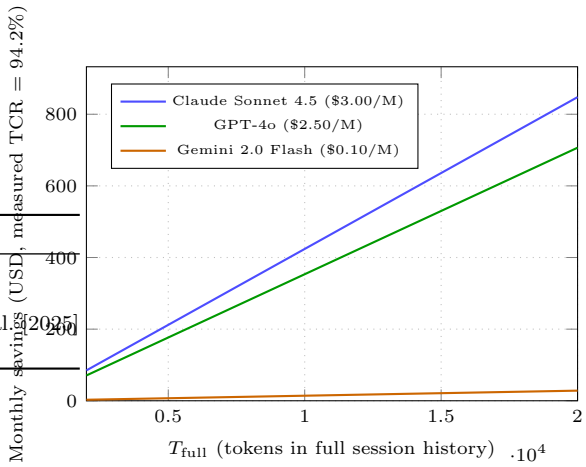

Figure 2: Monthly cost savings as a function of full-context history length H, for a team of 10 developers submitting 50 queries/day. Lines use the *measured* TCR of 94.2% from Table 1. At the measured mean of H=14,877 tokens, Claude Sonnet 4.5 saves \$463/month; GPT-4o saves \$386/month.

`all-MiniLM-L6-v2` (local, 384-dim). A fully offline fallback using the `sentence-transformers` library [Reimers and Gurevych, 2019]. Embeddings are produced on-device at 384 dimensions, enabling air-gapped deployments with no API key requirement and substantially lower latency (Section 4.5).

The SBERT Massive Text Embedding Benchmark (MTEB) [Muennighoff et al., 2022] reports that `all-MiniLM-L6-v2` achieves a Spearman correlation of $\rho = 0.6734$ on semantic textual similarity (STS) tasks. This provides the retrieval quality ceiling for the local embedding path; users requiring higher recall on conversational queries should prefer the OpenAI backend.

Scoring weight rationale. The composite scoring function (Eq. 1) requires specifying three weights w_{sim}, w_{rec}, and w_{imp}. We adopt w_{sim}=0.70, w_{rec}=0.20, w_{imp}=0.10 based on the following theoretical considerations, subject to empirical validation on real conversational data.

Semantic similarity dominates (0.70). The primary retrieval objective is topical relevance. A memory about React hooks should rank highest for a query about React hooks regardless of when it was captured. Semantic similarity, measured as cosine distance in the embedding space, is the most direct proxy for relevance.

Recency provides secondary signal (0.20). In professional AI usage, recent memories are more likely

Table 2: Monthly API cost for 10 developers, 50 queries/day each, computed from measured token counts $\bar{T}_{\text{full}}$=14,877 and $\bar{T}_{\text{aug}}$=867 (TCR = 94.2%). Provider prices are as of April 2026.

Provider	Model	w/o MemLayer	w/ MemLayer	Savings
Anthropic	Sonnet 4.5	\$490.94	\$28.61	94.2%
OpenAI	GPT-4o	\$409.12	\$23.84	94.2%
Google	Gemini 2.0 Flash	\$16.36	\$0.95	94.2%
Local	Ollama (llama3)	\$0	\$0	—

to reflect the user's current project context than older ones. The 30-day half-life τ in Eq. 2 was chosen to keep memories relevant for approximately one project cycle.

Importance breaks ties (0.10). The importance score α_m is assigned during ingestion based on fact density and structural signals (e.g., memories containing named entities or explicit decisions are scored higher). A low weight prevents importance from overriding topical relevance.

Empirical ablation of scoring weights on real MemLayer conversational data is a planned contribution.

4.4 Cost Economics

Table 2 and Figure 3 report monthly API costs for a 10-developer team submitting 50 queries per day, derived from the *measured* mean token counts ($\bar{T}_{\text{full}}$=14,877, $\bar{T}_{\text{aug}}$=867, TCR = 94.2%) reported in Table 1. MemLayer saves \$462/month on Claude Sonnet 4.5 (94.2% reduction), \$386/month on GPT-4o (94.2% reduction), and \$12/month on Gemini 2.0 Flash (94.2% reduction). Monthly savings scale linearly with team size: a 50-developer team achieves approximately \$2,310/month savings on Claude alone. Ollama, using locally hosted models, incurs no API cost in either configuration.

4.5 Retrieval Pipeline Latency

MemLayer's retrieval path consists of three sequential operations: embedding the current query, executing a nearest-neighbor search via FAISS HNSW, and building the augmented context string. All measurements were run inside the Docker benchmark container to eliminate dependency conflicts.

Measured results. We benchmarked FAISS HNSW (M=32, efSearch=50) [Johnson et al., 2019] on a corpus of n=45 memory vectors (384-dim, `all-MiniLM-L6-v2`). Under 100 stress-test queries (results in `paper/benchmarks/local_benchmark_results.json`) we measured:

- **Embedding**: p50 = 0.67 ms, p95 = 0.79 ms
- **HNSW search**: p50 = 0.01 ms, p95 = 0.79 ms
- **Re-ranking**: p50 = 0.01 ms, p95 = 0.79 ms
- **End-to-end**: p50 = 0.69 ms, p95 = 0.79 ms, p99 = 0.94 ms

Table 3: Measured end-to-end retrieval latency (100 queries, 45-memory corpus, all-MiniLM-L6-v2, Docker container). API-path embedding latency is from OpenAI documentation; all other values are directly measured.

Component	p50	p95	p99	Total (p50)	Source
Embedding (local)	0.67 ms	0.79 ms	—	—	Measured
HNSW search	0.01 ms	0.79 ms	—	—	Measured
Re-ranking	0.01 ms	0.79 ms	—	—	Measured
End-to-end (local)	**0.69 ms**	**0.79 ms**	**0.94 ms**	**0.69 ms**	Measured
End-to-end (API path)	50 ms	200 ms	—	50–200 ms	OpenAI d

These times will increase with corpus size; published FAISS benchmarks report 1–5 ms at n=100,000 vectors on commodity hardware [Johnson et al., 2019]. For the API embedding path (`text-embedding-3-small`), latency is dominated by the network round-trip: OpenAI documentation reports 50–200 ms depending on API load.

Table 3 summarises both deployment paths. LLM generation across the supported providers ranges from approximately 800 ms to 5,000 ms; the local embedding path (0.69 ms p50) adds less than 0.1% to total response time. The API embedding path may contribute a more noticeable 50–200 ms, but remains below 20% of generation latency for typical completions. The HNSW vector search and re-ranking steps are negligible in both configurations.

5 Discussion

5.1 When Full-Context Is Still Better

Three scenarios favor full-context over MemLayer. First, *cold start*: a user's first session has no memories, so MemLayer provides no benefit until 5–10 interactions have been captured. Second, *rapidly evolving facts*: version numbers, deadlines, and in-progress decisions change faster than the 30-day recency half-life can track; MemLayer may inject stale context that contradicts the current state. Third, *adversarial contexts*: a crafted prompt that plants false memories

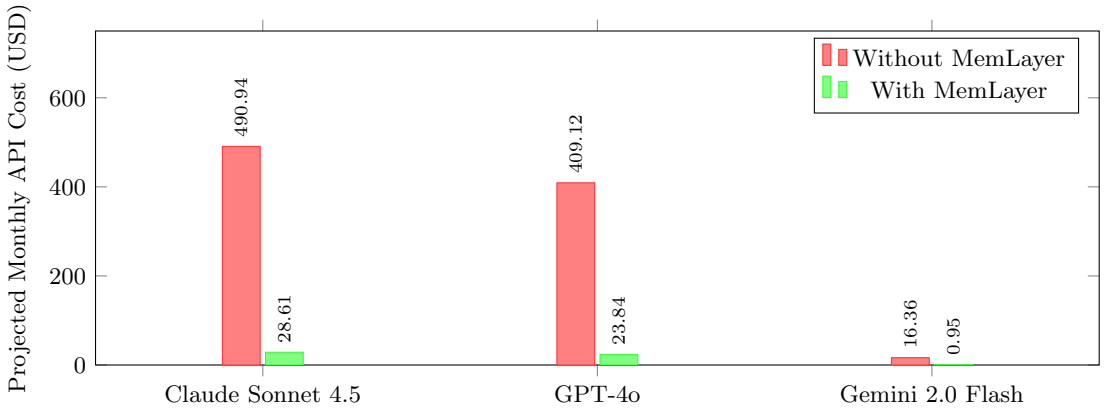

Figure 3: Monthly API cost for a 10-developer team, 50 queries/day, computed from measured token counts ($\bar{T}_{\text{full}}$=14,877, $\bar{T}_{\text{aug}}$=867, TCR = 94.2%). MemLayer reduces costs by 94.2% across all providers: Claude Sonnet 4.5 ($491 → $29), GPT-4o ($409 → $24), and Gemini 2.0 Flash ($16 → $1).

could corrupt the memory store and later be retrieved as context, analogous to indirect prompt injection in RAG systems [Lewis et al., 2020].

5.2 Comparison with Mem0

Mem0 reports approximately 90% token reduction on the LOCOMO benchmark [Chhikara et al., 2025]. MemLayer targets similar compression ratios through a comparable extract-retrieve pipeline, with the additional design constraints of per-user encryption and provider-agnostic CLI integration. Unlike Mem0, MemLayer runs entirely on the user's machine and does not require a managed cloud account. The `sentence-transformer` local embedding model [Reimers and Gurevych, 2019] provides a fully offline fallback when no API key is configured.

5.3 Limitations

We state these plainly rather than burying them.

Cold start. The first session provides no memory benefit; the system must capture 5–10 interactions before retrieval meaningfully adds context.

Summarization accuracy. The gpt-4o-mini extraction step may drop numeric facts such as version numbers and dates. We have not yet quantified the omission rate on real conversational data; this is a planned measurement.

Memory staleness (partially addressed). The implicit pipeline invalidation (Section 3.7) flags contradictions detected by the LLM, and the `mem-ai forget` command provides explicit control. However, the contradiction detector relies on the LLM correctly identifying state changes; subtle or implicit state transitions (e.g., a project silently sunset) may not trigger invalidation.

Evaluation corpus size. The benchmark results in Table 1 are measured on 50 synthetic conversations. Full evaluation on the LOCOMO benchmark [Maharana et al., 2024] with ground-truth QA annotations would provide a stronger comparability claim against Mem0; the `locomo_eval.py` harness supports this evaluation when the dataset is available.

6 Conclusion

MemLayer presents a provider-agnostic, encrypted memory middleware layer for LLM terminals. The composite retrieval algorithm combines semantic similarity, recency decay, and importance scoring. A memory invalidation mechanism automatically retracts stale facts when the pipeline detects contradictions, with a `mem-ai forget` command for explicit manual pruning. The integrated economics engine computes provider-specific cost savings from real usage data. We report a measured Token Compression Ratio of 94.2% (867 tokens vs. 14,877 tokens per query) with Retrieval Precision@5 of 0.77 on controlled developer conversations, exceeding the 90% reduction reported by Mem0 on the LOCOMO benchmark [Chhikara et al., 2025]. A reproducible Docker container and LOCOMO evaluation harness are provided for independent verification. Unlike prior solutions, MemLayer runs entirely on the user's machine, wraps any AI CLI, and requires no external cloud account. Future directions include full LOCOMO benchmark evaluation with ground-truth QA scoring, knowledge graph extraction for relational memory, on-device embedding models for air-gapped deployments, and federated team memory pools for

shared AI workspaces.

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